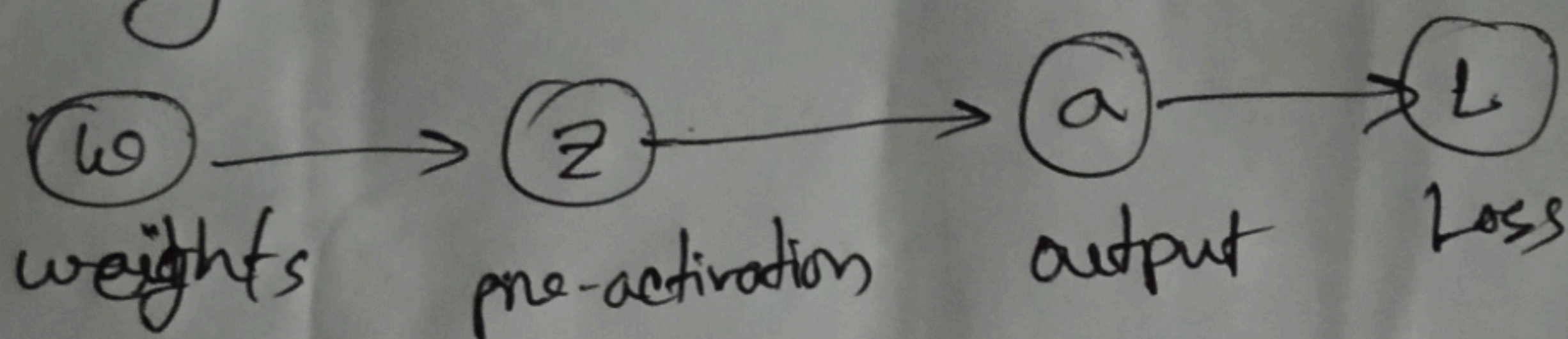


According to the chain rule,



and,

$$\Delta z = dz/dw \cdot \Delta w$$

$$\Delta a = da/dz \cdot \Delta z$$

$$\Delta L = dL/da \cdot \Delta a$$

$$\therefore \frac{dL}{dw} = \frac{dL}{da} \cdot \frac{da}{dz} \cdot \frac{dz}{dw}$$

then the sigmoid activation function is

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

differentiating with respect to z

$$\frac{d}{dz} \sigma(z) = \frac{e^{-z}}{(1+e^{-z})^2}$$

$$= \frac{1}{1+e^{-z}} \cdot \frac{e^{-z}}{1-e^{-z}}$$

$$= \sigma(z) (1 - \sigma(z))$$

$$= a(1-a)$$

and $d = (a-y)^2$

$$\therefore \frac{dL}{da} = 2(a-y)$$

Since $z = w \cdot x + b$

$$\frac{dz}{dw} = x$$

$$\therefore \frac{dL}{dw} = 2(a-y)a(1-a) \cdot x$$